

LiquidAI / LFM2.5-Encoder-350M-PII-Detector

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pii
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Use this model



Model card



Files



xet



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LFM2.5-Encoder-350-PII-Detector

A full fine-tune of [LFM2.5-Encoder-350M](#) with a token-classification head, covering **40 PII types** across **16 languages**.

Find more details about our encoders in our [blog post](#).

 **Demos:** Try this fine-tuned model running in a CPU-only Hugging Face space: [PII detection](#) — spot and remove 40 kinds of personal information across 16 languages.

Entity types (40 PII types across 11 domains)

Downloads last month
299



Safetensors ⓘ

Model size 0.4B params

Tensor type F32 [Files info](#)

Inference Providers **NEW**

Token Classification

This model isn't deployed by any Inference Provider.



Ask for provider support

Model tree for LiquidAI/LFM2.5-E...

Base model LiquidAI/LFM2.5-350M-Ba...



Finetuned LiquidAI/LFM2.5-En...


Finetuned (17) ... [this model](#)

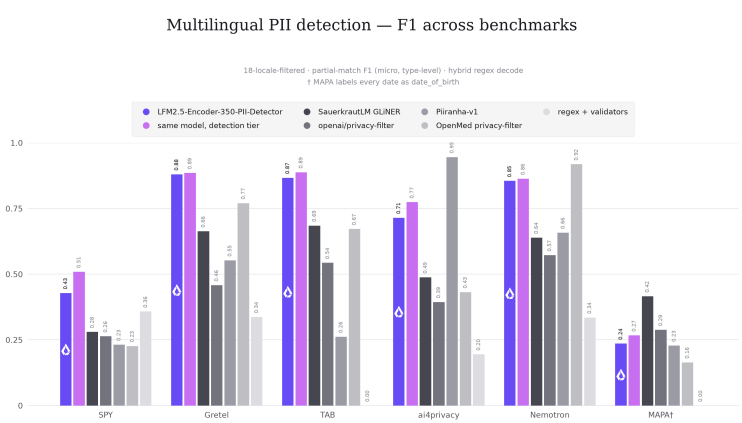
 Space using LiquidAI/LFM2.5-E... 1

Domain	Types
Identity	identity.person_name, identity.ssn, identity.national_id, identity.passport, identity.drivers_license, identity.date_of_birth, identity.tax_id
Contact	contact.email, contact.phone, contact.address, contact.postal_code, contact.ip_address
Financial	financial.credit_card, financial.iban, financial.bank_account, financial.swift_bic, financial.crypto_wallet, financial.amount
Credentials	credential.api_key, credential.password, credential.private_key, credential.jwt, credential.connection_string, developer.login_credentials
Online	online.username, online.url
Device	device.mac_address, device.imei, developer.device_id
Location	location.gps_coordinates
Healthcare	healthcare.medical_record, healthcare.condition, healthcare.medication, healthcare.health_plan_id

Domain	Types
Organization	org.company_name
Special-category	special.religion, special.political, special.orientation, special.health_status
Legal	legal.case_number

Benchmarks (18-locale-filtered, partial-F1, hybrid decode)

Benchmark	this model	detection tier	SauerkrautLM GLiNER	openAI GPT-4o
SPY	0.428	0.509	0.280	0.260
Gretel	0.880	0.885	0.663	0.450
TAB	0.867	0.888	0.685	0.540
ai4privacy	0.715	0.774	0.488	0.390
Nemotron	0.855	0.863	0.639	0.570
MAPA	0.236	0.267	0.416	0.280



- **Best on every benchmark except MAPA**, whose idiosyncratic date-as-date_of_birth labeling convention penalises correctly-typed predictions. Only two external scores land higher anywhere — Piirinha-v1's 0.946 on ai4privacy and OpenMed's 0.918 on Nemotron — and both are in-distribution: each model trains on that exact corpus, as did our own encoder's pretraining on those same two.
- **Detection tier** is the same model and the same predictions, scored with the type label ignored — did it find the PII span at all, which is the metric that matters for redaction. The gap to exact-type is the type-confusion rate, e.g. SPY 0.428 → 0.509.

Usage

“ *Loads custom code via `trust_remote_code=True` (the model wraps a `trust_remote_code` encoder).*”

Install the required packages:

```
pip install torch transformers huggingface_h
```

Run PII detection:

```
import importlib.util
import sys

from huggingface_hub import hf_hub_download
from transformers import AutoModelForTokenCl

model_id = "LiquidAI/LFM2.5-Encoder-350-PII-I
```

```

helper_path = hf_hub_download(model_id, "pii.
hf_hub_download(model_id, "context_cued.py")
sys.path.insert(0, helper_path.rsplit("/", 1)

spec = importlib.util.spec_from_file_location
hd = importlib.util.module_from_spec(spec)
spec.loader.exec_module(hd)

tok = AutoTokenizer.from_pretrained(model_id
model = AutoModelForTokenClassification.from

spans = hd.predict("Email Dr. Laura Schmidt ;
print(spans)

```

Contact

- Got questions or want to connect? [Join our Discord community](#).
- If you are interested in custom solutions with edge deployment, please contact [our sales team](#).

Citation

```

@article{liquidAI2026Encoders,
  author = {Liquid AI},
  title = {LFM2.5-Encoders: Fast at Long Con
  journal = {Liquid AI Blog},
  year = {2026},
  note = {www.liquid.ai/blog/lfm2-5-encoders}
}

```

